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### Key Points:

- Energetically based intensification rate equation predicts storms may reach peak intensity within 36 hr after reaching maximum intensification rate
- Time interval between peak intensification rate and peak intensity shortens as sea surface temperature increases in idealized full-physics numerical simulations
- From 1982 to 2023, the majority of global tropical cyclones peaked in intensity within 36 hr after reaching maximum intensification rate

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## The Intrinsic Timing Relationship Between Tropical Cyclone Maximum Intensification Rate and Lifetime Maximum Intensity

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**Abstract** The intensification rate (IR) equation, derived from a time-dependent simplified energetically based dynamical system (EBDS), links tropical cyclone (TC) intensity, IR, and their evolutions, suggesting a connection between the timing of TC lifetime maximum intensity (LMI) and the maximum intensification rate (MIR). This study demonstrates that the IR equation predicts MIR typically preceding LMI within 36 hr, as confirmed by numerical simulation and analysis of best track data. It was observed that the majority of global TCs reached LMIs within 36 hr after MIR from 1982 to 2023. This consistency between observational data, numerical simulations and analytical results highlights the EBDS-based IR equation's ability in capturing the timing relationship between MIR and LMI. This intrinsic link highlights the importance of understanding the timing of MIR in the evolution of TCs.

**Plain Language Summary** The damage caused by a tropical cyclone (TC) depends greatly on how strong it gets. One key measure of a TC's strength is its lifetime maximum intensity (LMI). While much research focuses on how intense a TC can become, less attention has been given to when it reaches its peak intensity. This study shows that the maximum intensification rate (MIR) usually happens within 36 hr before LMI, based on the intensification rate equation, which derives from a time-dependent simplified energetically based dynamical system. This finding is supported by the numerical simulation and analyzing the best track data. From 1982 to 2023, it was found that the majority of global TCs reached their LMIs within 36 hr after MIR. All observational data, numerical simulations and analytical analysis emphasize how important it is to understand the timing of MIR in the context of TC evolution.

## 1. Introduction

The mechanism behind tropical cyclone (TC) intensification has long been understood as positive feedback between the TC and the underlying ocean (Emanuel, 1988, 1989, 1995, 1997; Rotunno & Emanuel, 1987). Emanuel (1986, 1988) proposed the Carnot heat engine theory of TCs, which offers a quantifiable mathematical and physical framework for understanding potential intensity. According to this theory, TCs absorb energy from the ocean surface and release it in the cooler outflow layer, essentially operating like a Carnot heat engine. Within this theory, there exists an upper limit to TC intensity known as the maximum potential intensity (MPI), under favorable atmospheric and oceanic thermodynamic conditions (Emanuel, 1986, 1991, 1997).

Recent studies suggest that TCs also have a maximum potential intensification rate under given favorable conditions similar to MPI. Various equations describing the TC intensification rate (IR) over time have been developed to quantify TC IR. Emanuel (2012) formulated an IR equation based on the boundary layer entropy budget equation, assuming that a TC satisfies the axisymmetric thermal-wind equation derived from gradient wind balance and hydrostatic equilibrium. Ozawa and Shimokawa (2015) rederived the IR equation using the second law of thermodynamics based on the Carnot heat engine theory. Both time-dependent IR equations indicated that Maximum Intensification Rate (MIR) occurred at the beginning of intensification. However, this contradicted the observations indicating that MIR generally occurs at intermediate intensities rather than at the incipient stages (Xu & Wang, 2015).

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Wang, Li, Xu, Tan, and Lin (2021) introduced a concept called dynamical efficiency to the TC system and developed a simplified energetically based dynamical system (EBDS) model to quantify the intensity dependence of TC IR. Previous studies (Pendergrass & Willoughby, 2009; Schubert & Hack, 1982; Shapiro & Willoughby, 1982) have suggested that the heating efficiency of TC mainly depended on the inertia stability of the inner core, and the efficiency increases with the increase of the intensity. Thus, Wang, Li, Xu, Tan, and Lin (2021) defined dynamic efficiency as  $E = (I/I_{\text{MPI}})^n$ , where  $I$  is the inertia stability of inner core and  $I_{\text{MPI}}$  is the inertia stability of the TC when it reaches its MPI, unlike previous time-dependent theories that primarily consider environmental conditions (Demaria, 2009; Emanuel, 2012; Emanuel & Zhang, 2017; Ozawa & Shimokawa, 2015). The dynamical efficiency was subsequently parameterized as the normalized intensity, which was given as  $E = (V_m/V_{\text{MPI}})^{3/2}$ , where  $V_m$  is the maximum tangential wind,  $V_{\text{MPI}}$  is the MPI (Li et al., 2024; Wang, Li, & Xu, 2021). Subsequently, the simplified EBDS-based IR equation was formulated as follows:

$$\frac{\partial V_m}{\partial t} = \frac{C_D}{H} \left[ \left( \frac{V_m}{V_{\text{MPI}}} \right)^{\frac{3}{2}} V_{\text{MPI}}^2 - V_m^2 \right], \quad (1)$$

where,  $C_D$  represents the surface exchange coefficient of momentum (surface drag coefficient).  $H$  is roughly twice the depth of the boundary layer (Emanuel, 2012). Wang, Li, Xu, Tan, and Lin (2021) and Wang, Li, and Xu (2021) found that  $H = 2000$  m gives the best fit of the theoretical IR to the TC IR in their numerical simulations.

Equation 1 not only describes how IR changes over time but also reveals how IR varies with intensity. When the IR reaches its maximum value, denoted as MIR, its derivative with respect to intensity is 0, that is,  $\frac{\partial}{\partial V_m} \left( \frac{\partial V_m}{\partial t} \right) = 0$ . Therefore, MIR occurs at a specific intensity:

$$V_m = \frac{9}{16} V_{\text{MPI}}. \quad (2)$$

Wang, Li, and Xu (2021) validated Equation 2 by calculating theoretical IR distributions based on intensity and MPI (see their Figure 2), finding that their results were consistent with Equation 2. They further confirmed these theoretical calculations using best-track data (see their Figure 5), showing strong observational support. Xu and Wang (2022) provided additional validation by analyzing the EBDS model with observational data, further demonstrating its reliability. These findings suggest that IR peaks at an intermediate intensity, demonstrating that the EBDS model more effectively captures IR evolution compared to earlier formulations. Recent advancements have expanded the EBDS model's formulation and applicability. Wang et al. (2022, 2023) enhanced its precision by incorporating dissipative heating and isothermal expansion, making the theory both self-contained and adaptable to broader environmental conditions. Xu et al. (2023) integrated environmental factors into the model using machine learning to quantify their influence on TC intensification. Song et al. (2024) applied the refined theory to analyze long-term trends in TC IR over the western North Pacific, while Zhou et al. (2024) used the extended framework for TC intensity forecasting.

The EBDS model establishes a fundamental link between a storm's IR and intensity, indicating that TCs experience MIR when they reach 9/16 of their MPI. As the EBDS model is time-dependent, it implies an inherent relationship among intensity, IR, and their temporal evolution. Although Equation 1 does not provide a direct analytical solution for intensity over time, it hints at a relationship between the timing of MIR and the timing of a storm's peak intensity. To understand this relationship, we outline our analytical approach to derive a solution for the simplified EBDS-based IR equation and deduce expressions for the time interval between MIR and peak intensity. Furthermore, we conduct idealized full-physics numerical simulations to discuss the time interval between MIR and peak intensity under conditions involving more physical processes. Finally, we utilize best track data to validate the timing relationship between MIR and peak intensity for global TCs.

## 2. Solutions of IR Equation

Using dimensionless parameters is a common approach to tackle nonlinear dynamical problem. In this section, we simplify Equation 1 by introducing two dimensionless parameters. By solving the resulting dimensionless equation, we derive analytical expressions for the time interval  $\Delta t$  between MIR and peak intensity.

Xu & Wang (2022) demonstrated that while  $C_D$  and  $H$  are two independent parameters, in practice,  $C_D/H$  can be treated as a single parameter due to the difficulty in accurately determining either  $C_D$  or  $H$  from observations. Following the approach outlined by Xu & Wang (2022), we set  $C_D = 2.4 \times 10^{-3}$  and  $H = 2000$  m, and chose  $\kappa = C_D/H = 1.2 \times 10^{-6} \text{m}^{-1}$ . Consequently, Equation 1 can be reformulated as:

$$\frac{\partial V_m}{\partial t} = \kappa \left[ \left( \frac{V_m}{V_{\text{MPI}}} \right)^{\frac{3}{2}} V_{\text{MPI}}^2 - V_m^2 \right], \quad (3)$$

with the initial condition:

$$V_m(t=0) = V_0. \quad (4)$$

Here, we introduce two dimensionless parameters.  $t^*$  is termed as the dimensionless time, and  $V_m^*$  is termed as the dimensionless intensity.

$$t^* = \kappa V_{\text{MPI}} t, \quad (5)$$

$$V_m^* = \frac{V_m}{V_{\text{MPI}}}. \quad (6)$$

Substituting Equations 5 and 6 to Equations 3 and 4, we obtain the dimensionless IR equation and its initial condition:

$$\frac{\partial V_m^*}{\partial t^*} = \left( V_m^{*\frac{3}{2}} - V_m^{*2} \right), \quad (7)$$

$$V_0^*(t=0) = \frac{V_0}{V_{\text{MPI}}}. \quad (8)$$

We use variable substitution to solve this differential equation, and the solution given as follow:

$$\frac{t^*}{2} + \frac{1}{\sqrt{V_m^*}} + \ln \left( \frac{1 - \sqrt{V_m^*}}{\sqrt{V_m^*}} \right) = C(V_0^*), \quad (9)$$

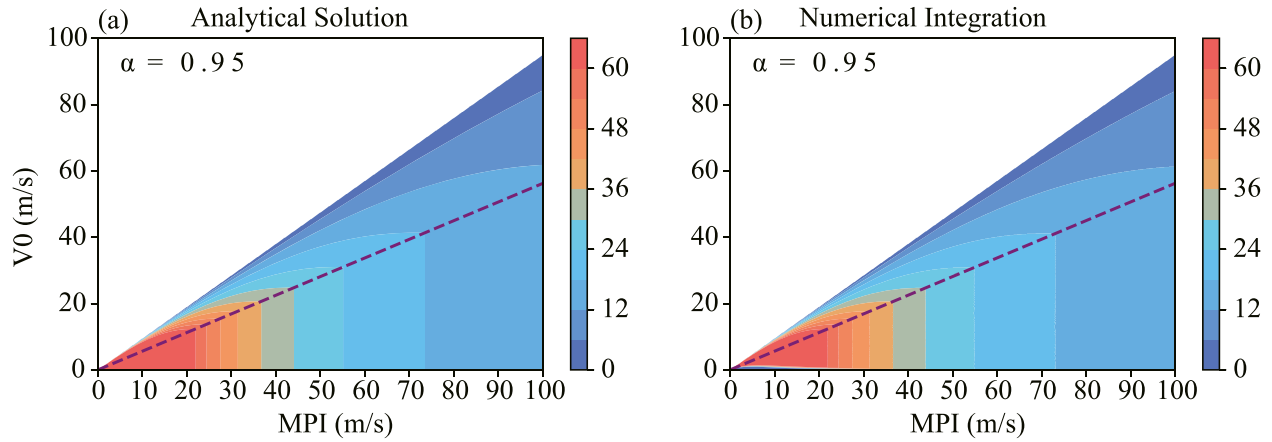
where  $C(V_0^*)$  is a parameter related to the initial intensity. Equation 9 provides the solution for both Equation 7 and the original Equation 3. For any given time  $t$ , there is a corresponding intensity  $V_m$ . Equation 9 quantifies the time dependence of intensity, allowing us to calculate the intrinsic timing relationships between different intensities.

When a TC reaches a certain intensity  $V_{\text{MIR}} = 9/16 V_{\text{MPI}}$ , it experiences its MIR as described by Wang, Li, and Xu (2021). After  $V_{\text{MIR}}$ , the TC's intensity continues to increase, but at a reduced IR until the end of the intensification period. We define  $V_S = \alpha V_{\text{MPI}}$  as steady-state intensity, which approximates when the storm reaches its maximum intensity. We denote the moment of MIR as  $t_1$ , and the moment of steady-state intensity as  $t_2$ . By substituting  $V_{\text{MIR}} = 9/16 V_{\text{MPI}}$  and  $V_S = \alpha V_{\text{MPI}}$  into Equation 9, we obtain:

$$t_1^* = 2 \left( C - \frac{4}{3} + \ln 3 \right), \quad (10)$$

$$t_2^* = 2 \left( C - \frac{1}{\sqrt{\alpha}} - \ln \frac{1 - \sqrt{\alpha}}{\sqrt{\alpha}} \right). \quad (11)$$

The intensification process is governed by the initial intensity  $V_0$  and the critical intensity  $V_{\text{MIR}}$ . If the initial intensity at which a storm begins to intensify is weaker than the threshold value, that is,  $V_0 < V_{\text{MIR}}$ , meaning the storm has not yet reached its MIR, we can calculate dimensionless time interval as:



**Figure 1.** Time interval  $\Delta t$  as a function of initial intensity and MPI for  $\alpha = 0.95$  in (a) the analytical solution, and (b) the numerical integration solution. The purple dashed line represents the intensity of  $V_0 = V_{\text{MIR}}$ . The shading represents different time intervals.

$$\Delta t^* = t_2^* - t_1^* = 2 \left( \frac{4}{3} - \frac{1}{\sqrt{\alpha}} + \ln \frac{\sqrt{\alpha}}{3(1-\sqrt{\alpha})} \right). \quad (12)$$

The dimensionless time interval is a function of  $\alpha$  only. For given parameters  $\alpha$  and  $\kappa$ , the time interval is inversely proportional to MPI:

$$\Delta t = \frac{\Delta t^*}{\kappa V_{\text{MPI}}}. \quad (13)$$

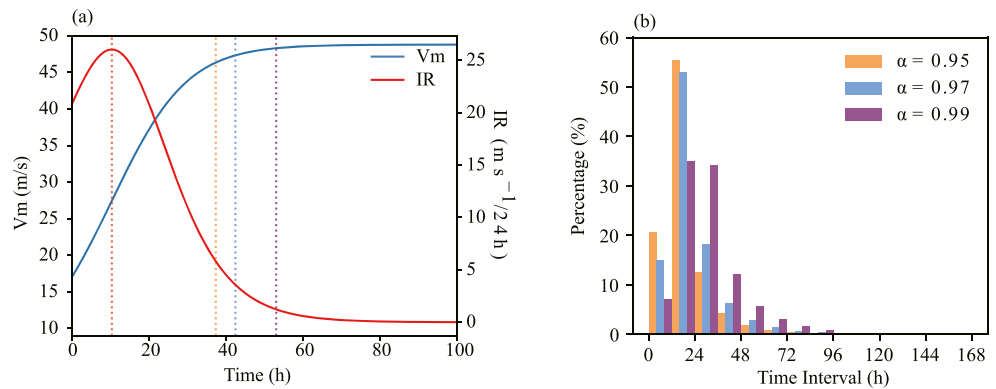
If the initial intensity exceeds the critical value, that is,  $V_0 \geq V_{\text{MIR}}$ , the TC undergoes its MIR as it begins to intensify, meaning  $t_1 = 0$  and  $t_1^* = 0$ , resulting a time interval of:

$$\Delta t = \frac{t_2^*}{\kappa V_{\text{MPI}}}. \quad (14)$$

In this case, the time interval is also inversely proportional to MPI.

We derive general analytical expressions for the time interval between MIR and steady-state intensity as shown in Equations 13 and 14. To validate these expressions, we performed numerical integration of Equation 3 using the same parameters. The time intervals for different initial intensities when the MPI ranges from 0 to 100 m/s based on analytical solution and numerical integration are shown in Figures 1a and 1b, respectively. The patterns in Figures 1a and 1b are nearly identical, with minor discrepancies occurring when  $V_0$  is close to 0 and MPI is small. In these cases, the storm's lifetime is very long, exceeding the 200-hr limit of our numerical integration, leading to these differences. In our numerical setup, where  $V_0$  and MPI range from 0 to 100 m/s, the storm's lifetime typically does not exceed 200 hr. The agreement between analytical solution and numerical integration confirms the validity of the analytical expression for the time interval. Both Figures 1a and 1b show that the time interval decreases monotonically as MPI increases and is independent of the initial intensity, when  $V_0 < V_{\text{MIR}}$ . The intensity difference between the MIR and MPI is 7/16 of MPI, meaning that as MPI increases, the intensity difference between MIR and MPI also grows. This effect tends to increase the time interval. However, an increase in MPI also leads to a higher IR, which tends to shorten the time interval. These two competing effects interact, but the overall inverse relationship between the time interval and MPI suggests that the increase in IR dominates the effect of the growing intensity difference. However, when  $V_0 \geq V_{\text{MIR}}$ , the time interval depends on both the initial intensity and MPI.

In the analytical solution, the timing of the storm reaching its peak intensity is a function of  $\alpha$ , which is represented by  $V_S = \alpha V_{\text{MPI}}$ . In the IR equation, the  $\alpha$  should be close to, but less than, 1. In real and simulated TCs, peak intensity can be equal or greater than  $V_{\text{MPI}}$ , which is often termed “super intensity” (Li et al., 2020; Persing &



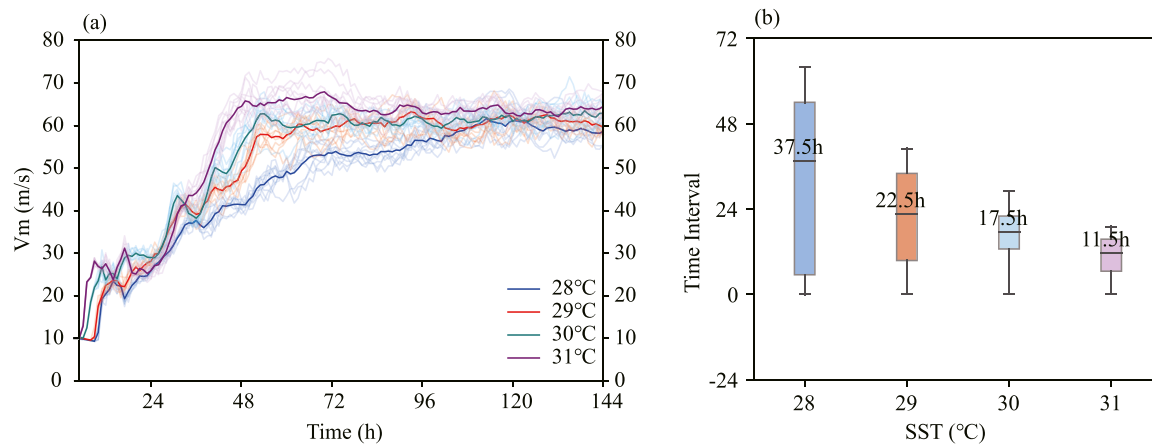
**Figure 2.** (a) A solution of the IR equation with the initial intensity set to 17 m/s and MPI set to 49 m/s. The red dash line represents  $t_1$ , the orange dashed line, blue dashed line, and purple dashed line represent  $t_2$  correspond to  $\alpha$  values of 0.95, 0.97, and 0.99, respectively; (b) Percentage of different time intervals in Figure 1a for various  $\alpha$ .

Montgomery, 2003). Figure 2a shows a solution to Equation 3 as an example demonstrating how the value of  $\alpha$  influence time interval, with  $V_0 = 17$  m/s and  $V_{MPI} = 49$  m/s. Different colored dash lines represent the timing of peak intensity for various values of  $\alpha$ . When  $\alpha$  is less than 0.95, the intensification process is not yet complete, while  $\alpha$  greater than 0.99 indicates that no further intensification occurs. Therefore, the range of  $\alpha$  between 0.95 and 0.99 is considered reasonable for estimating the time interval. To demonstrate the impact of different  $\alpha$  values on the analytical results, Figure 2 shows the percentages of time interval from Figure 1a. For  $\alpha = 0.95$ , 91.8% of time intervals are within 36 hr, with 57.4% falling between 12 and 24 hr. Similarly, for  $\alpha = 0.97$ , 87.7% of intervals are within 36 hr, with 53.8% in the 12–24 hr range. In the case of  $\alpha = 0.99$ , 76.3% of intervals remain within 36 hr, with the highest likelihood (34.9%) between 12 and 24 hr. As  $\alpha$  increases, the time interval also increases because higher  $\alpha$  results in a larger intensity difference between  $t_1$  and  $t_2$  for the same MPI. These results suggest that while  $\alpha$  values close to 1 affect the distribution of time intervals, they have a relatively small impact on the overall range, with most time intervals staying within 36 hr. This indicates that the EBDs-based IR equation predicts storms may reach peak intensity within 36 hr after achieving their MIR.

### 3. Idealized Full-Physics Numerical Simulations

To determine whether the timing relationship between MIR and peak intensity holds under more complex physical conditions, a series of idealized full-physics numerical simulations were conducted. The simulations were performed using the idealized TC case in the Weather Research and Forecasting (WRF) model, version 3.7.1 (Skamarock et al., 2008). The model domain covers an area of 2,000 km  $\times$  2,000 km, with a horizontal grid spacing of 4 km, and includes 31 vertical levels in the atmosphere. An f-plane is assumed  $f = 5 \times 10^{-5} \text{ s}^{-1}$ , corresponding to 20°N. The cloud microphysics are handled using the Thompson et al. (2008) scheme, with cumulus convection omitted following Li et al. (2020). The Rapid Radiative Transfer Model (RRTM) scheme (Mlawer et al., 1997) is used for longwave radiation transfer processes. The Dudhia (1989) scheme is used for shortwave radiation transfer processes. The planetary boundary layer is represented by the YSU scheme (Hong et al., 2006), and surface-layer processes are modeled using the Revised MM5 Monin-Obukhov scheme (Jimenez et al., 2012). The ocean mixed layer scheme is turned off, so sea surface temperatures (SSTs) remain fixed throughout the simulations. The vortex is in hydrostatic and gradient-wind balance, defined using the analytic formulation from Rotunno and Emanuel (1987). The thermodynamic initial conditions follow the Jordan (1958) sounding. The surface enthalpy flux exchange coefficient is constant, while the surface drag coefficient is based on Donelan et al. (2004).

We ran four ensemble experiments with varying SSTs, ranging from 28°C to 31°C, each comprising 10 runs. To introduce variability, the runs were initialized on different start dates, making them under slightly different solar radiation conditions. The initial intensity of the vortex was set at 15 m/s, with an initial radius of maximum wind (RMW) of 82.5 km. Small perturbations were applied to the initial intensity and size of the storm to the runs to further enhance variability. All simulations were integrated over a 7-day period. Model outputs were generated hourly, and the IR was calculated as the 6-hr change in maximum wind speed,  $IR_{6h} = (V_m^{t+6h} - V_m^t)/6h$ . The



**Figure 3.** (a) The time evolution of TC intensity simulated by the models, with the thin lines showing individual runs for each ensemble group and the solid lines representing the ensemble mean; (b) Box plots of time interval between MIR and peak intensity for each run.

timing of MIR and the first occurrence of peak intensity were recorded. To ensure the time interval was not influenced by the choice of the IR period, if a TC reached peak intensity immediately after MIR, the time interval between MIR and peak intensity was considered to be 0 hr.

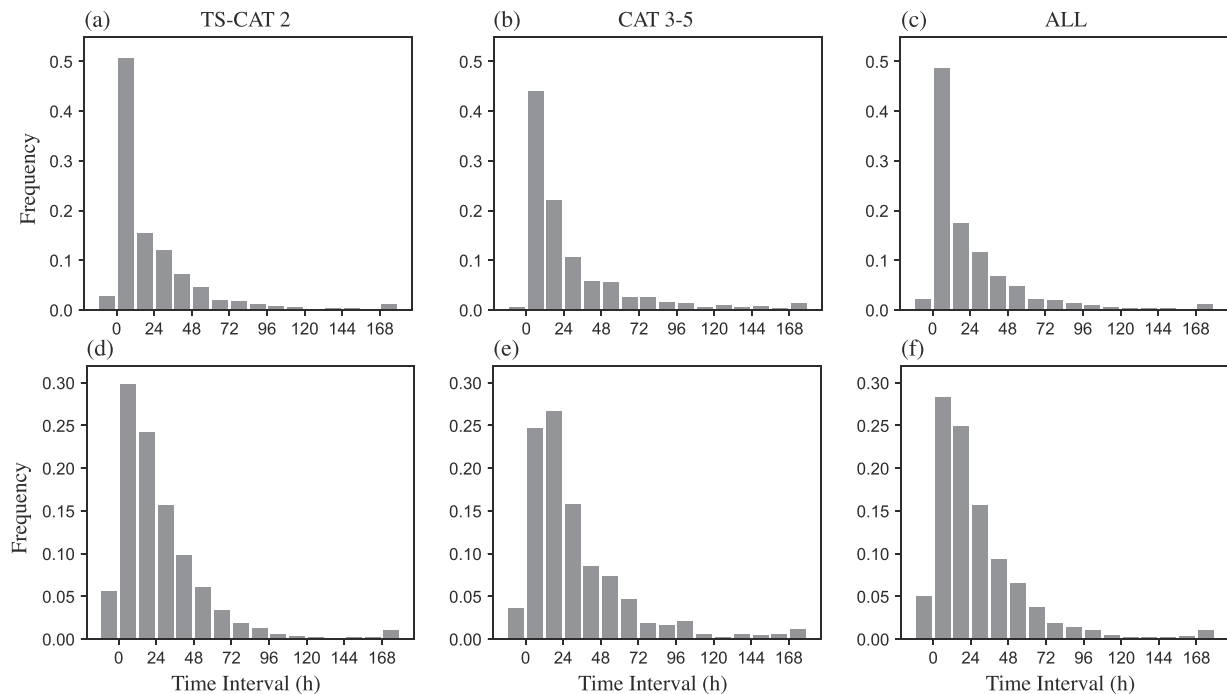
Figure 3a displays the intensity evolution for each run and the ensemble mean in the four experiments. Higher SSTs led to greater peak intensities and earlier achievement of these intensities, as seen in the ensemble means. For each simulation, we calculated the time interval between MIR and peak intensity. Figure 3b shows the distribution of these intervals across the four ensemble experiments. Notably, 80.0% of the simulated TCs reached peak intensity within 36 hr after completing MIR, closely matching the distribution predicted by theoretical models. The median time intervals for the four SST groups were 37.5, 22.5, 17.5, and 11.5 hr, respectively. As SST increased (indicating higher MPI), the medians and upper quartiles of the time intervals shortened. This is consistent with analytical solutions in Equations 13 and 14, which show that the time interval between MIR and peak intensity decreases as MPI increases.

#### 4. Evaluations Based on Observation Data

Both the analytical solutions and idealized full-physics numerical simulations suggest most storms reach their peak intensity within 36 hr of completing MIR. In this section, we examine real-world scenarios using observational data to validate this phenomenon. We utilize the International Best Track Archive for Climate Stewardship data set (IBTrACS; Knapp et al., 2018), which covers global TC parameters from 1982 to 2023. This data set provides measurements of TC maximum 1-min mean sustained surface winds every 6 hr. We focus on storms with tropical storm (TS) level or higher. If a storm experiences multiple occurrences of lifetime maximum intensity (LMI) throughout its lifetime, we only consider the first LMI to focus on the primary intensification period.

Figure 4 displays the frequency distribution of the time interval between MIR and the moment when storms reach their LMIs for both weak and strong storms in the global ocean from 1982 to 2023. During this period, 2,232 weak storms with intensities ranging from TS to Category 2 and 989 strong storms with intensity above Category 3 are analyzed. In Figures 4a–4c, if a storm experienced multiple occurrences of MIR, only the last MIR event was considered. The method for calculating the time interval between MIR and LMI is the same as described in Section 3. Among weak storms, 50.6% reached their LMIs within 12 hr after experiencing MIR, while 77.9% reached their LMIs within 36 hr after experiencing MIR. Similarly, among strong storms, 44.0% reached their LMIs 12 hr after experiencing MIR, while 76.5% reached their LMIs within 36 hr after experiencing MIR. Time intervals between MIR and LMI were comparable for both weak and strong storms. Considering all global TCs combined, 48.5% reached their LMIs 12 hr after experiencing MIR, while 77.5% reached their LMIs within 36 hr after experiencing MIR. A small percentage of storms (2.82% for weak storms and 0.40% for strong storms) experienced their LMIs before MIR, indicating a cycle of intensification, weakening, and subsequent intensification, possibly related to eyewall replacement cycles (Sitkowski et al., 2011). Storms with longer intervals





**Figure 4.** Frequency distribution of time interval between MIR and LMI for global TCs from 1982 to 2023. (left) TS-CAT 2, (middle) CAT 3–5, and (right) all TCs. Panels (a–c) define the timing of MIR as the endpoint of the IR period, while panels (d–f) define MIR timing as the average time of the IR period. For storms with multiple MIR events, the MIR timing in panels (d–f) is calculated as the average of all such events. LMI occurrences before MIR and those occurring 168 hr after MIR are grouped into the first and last bars, respectively.

between MIR and LMI often experienced a temporary weakening after MIR, followed by secondary intensification before reaching LMI. This pattern could result from rapidly changing MPI or unfavorable environmental conditions after MIR. The theoretical time interval derived from Equation 1 does not include eyewall replacement cycles or environmental variability, limiting its ability to explain real-world multi-intensification events. Despite this limitation, the observed time interval between MIR and LMI align remarkably well with the theoretical results, indicating the simplified EBDS-based IR equation's ability in capturing the primary intensification dynamics.

In Figures 4d–4f, the timing of MIR was defined as the average time of the IR period. For storms with multiple MIR events, the timing was calculated as the average of all such events. Under this definition, 54.0% of weak storms reached their LMIs within 24 hr after MIR, and 69.6% within 36 hr. Similarly, 51.4% of strong storms reached their LMIs within 24 hr after MIR, and 67.1% within 36 hr. For all global TCs combined, 53.2% reached their LMIs within 24 hr after MIR, while 68.9% did so within 36 hr. Regardless of how MIR timing is defined, the majority of global TCs reached their LMIs within 36 hr after MIR.

Overall, the observational data align with the theoretical results obtained from the IR equation, indicating that the majority of storms reach their LMIs within 36 hr after experiencing MIR. Since the timing of LMI is approximated from the timing of the MPI in the theoretical analysis, the consistent time intervals between MIR and LMI from the observational data and the theoretical analysis indicate that the timing of LMI can be well represented by the timing of the MPI, even though the majority of storms could only achieve approximately 55% of their MPIs at their LMIs (DeMaria & Kaplan, 1994; Wang & Wu, 2024).

## 5. Summary and Discussion

Recent studies (Li et al., 2024; Wang, Li, & Xu, 2021; Wang, Li, Xu, Tan, & Lin, 2021; Xu & Wang, 2022) have proposed a time-dependent EBDS-based IR equation, replicating TC IR dependence on TC intensity. This equation establishes a connection between TC intensity, IR, and their evolution, implying a relationship between the timing of TC peak intensity and the occurrence of MIR. The analysis based on EBDS-based IR equation demonstrates that the time interval is inversely proportional to MPI. Idealized full-physics numerical simulations

show that as SST increases, and therefore MPI, the time interval for TCs to reach their peak intensity shortens. These simulations confirm that the timing relationship between MIR and steady-state intensity predicted by the theoretical model holds true in more complex numerical models.

The EBDS-based IR equation predicts that MIR occurs within 36 hr before steady-state intensity is reached, a result supported by both numerical simulations and best track data. 80.0% of the simulated time intervals fall within this 36-hr range for SSTs between 28°C and 31°C, closely matching theoretical predictions. From 1982 to 2023, observational data show that majority of global TCs reached their LMI within 36 hr. The consistency between observations and theoretical analysis highlights the equation's ability to capture the timing relationship between MIR and LMI. This suggests that even though most storms only achieve about 55% of their MPI at LMI, the timing of LMI is closely related to the timing of MPI (DeMaria & Kaplan, 1994; Wang & Wu, 2024).

This study employs a simplified EBDS model to investigate the intrinsic timescale between MIR and LMI. While the model excludes processes such as dissipative heating (Wang et al., 2022), isothermal expansion (Wang et al., 2023), environmental influences (Xu et al., 2023), and eyewall replacement cycles, these omissions do not compromise its primary aim, which is to establish the conceptual foundation of a MIR-LMI timescale rather than replicate precise observational values. Despite its simplifications, the model successfully reproduces the observed MIR-LMI timescale, indicating its ability to capture the fundamental processes of intensification.

While much research has focused on how strong a TC can become, less attention has been given to when it will reach its peak intensity. Recent studies have begun to address this gap. Li et al. (2022) examined the timing relationship between RMW contraction and intensification. They found that the end of RMW contraction preceded the end of intensification in more than half of hurricanes, with the preceding time typically within 48 hr. This study highlights the intrinsic link between the timing of LMI and MIR, demonstrating the importance of understanding MIR timing in the evolution of TCs through analytical models, numerical simulations, and observational data. Since the end of intensification marks the attainment of LMI, and given the similarities between these processes, future research could investigate how the timing of MIR is linked to the end of RMW contraction and explore the underlying dynamics driving this relationship.

## Data Availability Statement

The best-track data were obtained from Knapp et al. (2018) and available at <https://www.ncei.noaa.gov/data/international-best-track-archive-for-climate-stewardship-ibtracs/v04r01/access/csv/>.

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